Chapter 7: Duality

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The Primal and Dual Problems

Consider the problem

$$\begin{aligned} f^* := \min\ & f(\mathbf{x}) \\ \text{s.t.}\quad & g_i(\mathbf{x}) \leq 0, i = 1, 2, \ldots, m \\ & h_j(\mathbf{x}) = 0, j = 1, 2, \ldots, p, \\ & \mathbf{x} \in X\,, \end{aligned} \tag{Primal}$$

where $f, g_i, h_j (i = 1, 2, \ldots, m, j = 1, 2, \ldots, p)$ are functions defined on the set $X \subseteq \mathbb{R}^n$. This is the "usual" optimization problem, and we will refer to it as the **primal** problem. As discussed in last week, the Lagrangian associated to this problem is

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = f(\mathbf{x}) + \sum_{i=1}^{m} \lambda_i g_i(\mathbf{x}) + \sum_{j=1}^{p} \mu_j h_j(\mathbf{x}) \quad \left(\mathbf{x} \in X, \boldsymbol{\lambda} \in \mathbb{R}^m_+, \boldsymbol{\mu} \in \mathbb{R}^p\right) .$$

The **dual** objective function $q : \mathbb{R}^m_+ \times \mathbb{R}^p \to \mathbb{R} \cup \{-\infty\}$ is defined to be

$$q(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \min_{\mathbf{x} \in X} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$$

The domain of the dual objective function is

$$\operatorname{dom}(q) = \left\{(\boldsymbol{\lambda}, \boldsymbol{\mu}) \in \mathbb{R}^m_+ \times \mathbb{R}^p : q(\boldsymbol{\lambda}, \boldsymbol{\mu}) > -\infty\right\} .$$

The **dual problem** is given by

$$\begin{aligned} q^* :=\ & \max\ q(\boldsymbol{\lambda}, \boldsymbol{\mu}) \\ \text{s.t.}\quad & (\boldsymbol{\lambda}, \boldsymbol{\mu}) \in \operatorname{dom}(q) \end{aligned} \tag{Dual}$$

In many optimization problems it is useful to study the properties of the dual problem, and even resorting to solving the dual problem instead. This is an open-ended question that we shall explore in this week, trying to understand when and why is this good idea, and illustrating with some relevant examples. We begin by stating some relevant properties of the dual problem.

Theorem. *Consider the primal problem* (Primal) *with* $f, g_i, h_j (i = 1, 2, \dots, m,\ j = 1, 2, \dots, p)$ *being functions defined on the set* $X \subseteq \mathbb{R}^n$*, and let* q *be the dual function defined in* (Dual)*. Then:*

a) $\operatorname{dom}(q)$ *is a convex set.*

b) q *is a concave function over* $\operatorname{dom}(q)$.

Proof. (a) Take $\left(\boldsymbol{\lambda}_1, \boldsymbol{\mu}_1\right), \left(\boldsymbol{\lambda}_2, \boldsymbol{\mu}_2\right) \in \operatorname{dom}(q)$ and $\alpha \in [0, 1]$. Then

$$\begin{aligned} \min_{\mathbf{x} \in X} L\left(\mathbf{x}, \boldsymbol{\lambda}_1, \boldsymbol{\mu}_1\right) &> -\infty\,, \\ \min_{\mathbf{x} \in X} L\left(\mathbf{x}, \boldsymbol{\lambda}_2, \boldsymbol{\mu}_2\right) &> -\infty\,. \end{aligned}$$

Therefore, since the Lagrangian $L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$ is affine w.r.t. $\boldsymbol{\lambda}, \boldsymbol{\mu}$

$$\begin{aligned} q\left(\alpha\boldsymbol{\lambda}_1 + (1-\alpha)\boldsymbol{\lambda}_2, \alpha\boldsymbol{\mu}_1 + (1-\alpha)\boldsymbol{\mu}_2\right) &= \min_{\mathbf{x} \in X} L\left(\mathbf{x}, \alpha\boldsymbol{\lambda}_1 + (1-\alpha)\boldsymbol{\lambda}_2, \alpha\boldsymbol{\mu}_1 + (1-\alpha)\boldsymbol{\mu}_2\right) \\ &= \min_{\mathbf{x} \in X} \left\{\alpha L\left(\mathbf{x}, \boldsymbol{\lambda}_1, \boldsymbol{\mu}_1\right) + (1-\alpha) L\left(\mathbf{x}, \boldsymbol{\lambda}_2, \boldsymbol{\mu}_2\right)\right\} \\ &\geq \alpha \min_{\mathbf{x} \in X} L\left(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}_1\right) + (1-\alpha) \min_{\mathbf{x} \in X} L\left(\mathbf{x}, \boldsymbol{\lambda}_2, \boldsymbol{\mu}_2\right) \\ &= \alpha q\left(\boldsymbol{\lambda}_1, \boldsymbol{\mu}_1\right) + (1-\alpha) q\left(\boldsymbol{\lambda}_2, \boldsymbol{\mu}_2\right) \\ &> -\infty \end{aligned}$$

Hence, $\alpha\left(\boldsymbol{\lambda}_1, \boldsymbol{\mu}_1\right) + (1-\alpha)\left(\boldsymbol{\lambda}_2, \boldsymbol{\mu}_2\right) \in \operatorname{dom}(q)$, and the convexity of $\operatorname{dom}(q)$ is established. (b) $L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$ is an affine function w.r.t. $(\boldsymbol{\lambda}, \boldsymbol{\mu})$. In particular, it is a concave function w.r.t. $(\boldsymbol{\lambda}, \boldsymbol{\mu})$. Hence, since q is the minimum of concave functions, it must be concave. □

Weak and Strong Duality

A first important consequence for optimization is the week duality theorem, which establishes a lower bound for the primal optimal value with respect to the dual optimal value.

Theorem (Weak Duality Theorem). *Consider the primal problem* (Primal) *and its dual problem* (Dual). *Then*

$$q^* \leq f^*$$

where f^*, q^* are the primal and dual optimal values respectively.

Proof. The feasible set of the primal problem is

$$S = \left\{ \mathbf{x} \in X : g_i(\mathbf{x}) \leq 0, h_j(\mathbf{x}) = 0, i = 1, 2, \dots, m, j = 1, 2, \dots, p \right\} .$$

Then for any $(\boldsymbol{\lambda}, \boldsymbol{\mu}) \in \mathrm{dom}(q)$ we have

$$\begin{aligned} q(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \min_{\mathbf{x} \in X} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) &\leq \min_{\mathbf{x} \in S} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) \\ &= \min_{\mathbf{x} \in S} \left\{ f(\mathbf{x}) + \sum_{i=1}^{m} \lambda_i g_i(\mathbf{x}) + \sum_{j=1}^{p} \mu_j h_j(\mathbf{x}) \right\} \\ &\leq \min_{\mathbf{x} \in S} f(\mathbf{x}) = f^* . \end{aligned}$$

Taking the maximum over $(\boldsymbol{\lambda}, \boldsymbol{\mu}) \in \mathrm{dom}(q)$, the result follows. □

Example:

$$\begin{array}{ll} \min & x_1^2 - 3x_2^2 \\ \text{s.t.} & x_1 = x_2^3 \end{array}$$

While the weak duality theorem is useful to obtain a lower bound for the optimal value of the primal problem, a more powerful result can be proven known as **strong duality**. For this, we need to cast a nonlinear variant of Farkas' Lemma, which we briefly state.

Theorem (Supporting Hyperplane Theorem). *Let $C \subseteq \mathbb{R}^n$ be a convex set and let $\mathbf{y} \notin C$. Then there exists $\mathbf{0} \neq \mathbf{p} \in \mathbb{R}^n$ such that*

$$\mathbf{p}^T \mathbf{x} \leq \mathbf{p}^T \mathbf{y} \text{ for any } \mathbf{x} \in C .$$

Theorem (Separation of Two Convex Sets). *Let $C_1, C_2 \subseteq \mathbb{R}^n$ be two nonempty convex sets such that $C_1 \cap C_2 = \emptyset$. Then there exists $\mathbf{0} \neq \mathbf{p} \in \mathbb{R}^n$ for which*

$$\mathbf{p}^T \mathbf{x} \leq \mathbf{p}^T \mathbf{y} \text{ for any } \mathbf{x} \in C_1, \mathbf{y} \in C_2 .$$

Theorem (Nonlinear Farkas Lemma). *Let $X \subseteq \mathbb{R}^n$ be a convex set and let $f, g_1, g_2, \dots, g_m$ be convex functions over X. Assume that there exists $\hat{x} \in X$ such that*

$$g_1(\hat{\mathbf{x}}) < 0, g_2(\hat{\mathbf{x}}) < 0, \dots, g_m(\hat{\mathbf{x}}) < 0 .$$

Let $c \in \mathbb{R}$. Then the following two claims are equivalent:

a) The following implication holds:

$$\mathbf{x} \in X, g_i(\mathbf{x}) \leq 0, i = 1, 2, \ldots, m \Rightarrow f(\mathbf{x}) \geq c\,.$$

b) There exist $\lambda_1, \lambda_2, \ldots, \lambda_m \geq 0$ such that

$$\min_{x \in X} \left\{ f(x) + \sum_{i=1}^{m} \lambda_i g_i(x) \right\} \geq c\,.$$

With these technical results, we are in position to state the strong duality result.

Theorem (Strong Duality of Convex Problems with Inequality Constraints)**.** *Consider the optimization problem*

$$\begin{aligned} f^* = &\min f(\mathbf{x}) \\ \text{s.t.}\ \ &g_i(\mathbf{x}) \leq 0, \quad i = 1, 2, \ldots, m, \\ &\mathbf{x} \in X \end{aligned}$$

where X is a convex set and $f, g_i, i = 1, 2, \ldots, m$ are convex functions over X. Suppose that there exists $\hat{\mathbf{x}} \in X$ for which $g_i(\hat{\mathbf{x}}) < 0, i = 1, 2, \ldots, m$. If this problem has a finite optimal value, then

a) the optimal value of the dual problem is attained.

b) the primal and dual problems have the same optimal value, $f^* = q^*$.

Proof. Since $f^* > -\infty$ is the optimal value of the primal problem, it follows that the following implication holds:

$$\mathbf{x} \in X, g_i(\mathbf{x}) \leq 0, i = 1, 2, \ldots, m \Rightarrow f(\mathbf{x}) \geq f^*\,.$$

By the nonlinear Farkas Lemma, there exists $\tilde{\lambda}_1, \tilde{\lambda}_2, \ldots, \tilde{\lambda}_m \geq 0$ such that

$$q(\tilde{\lambda}) = \min_{\mathbf{x} \in X} \left\{ f(\mathbf{x}) + \sum_{j=1}^{m} \tilde{\lambda}_j g_j(\mathbf{x}) \right\} \geq f^*\,.$$

By the weak duality theorem,

$$q^* \geq q(\tilde{\boldsymbol{\lambda}}) \geq f^* \geq q^*\,.$$

Hence, $f^* = q^*$ and $\tilde{\boldsymbol{\lambda}}$ is an optimal solution of the dual problem. □

The result above indicates that under the convexity assumptions of the theorem, it is possible to obtain the solution of the primal problem by solving its dual.

Example:

$$\begin{aligned} \min \quad & x_1^2 - x_2 \\ \textbf{s.t.} \quad & x_2^2 \leq 0\,. \end{aligned}$$

Theorem (Complementary Slackness Conditions)**.** *Consider the optimization problem*

$$f^* := \min\left\{f(\mathbf{x}) : g_i(\mathbf{x}) \leq 0, i = 1, 2, \ldots, m,\ \mathbf{x} \in X\right\}\,,$$

and assume that $f^* = q^*$ where q^* is the optimal value of the dual problem. Let $\mathbf{x}^*, \boldsymbol{\lambda}^*$ be feasible solutions of the primal and dual problems. Then $\mathbf{x}^*, \boldsymbol{\lambda}^*$ are optimal solutions of the primal and dual problems iff

$$\begin{gathered} \mathbf{x}^* \in \operatorname*{argmin}_{\mathbf{x} \in X} L\left(\mathbf{x}, \boldsymbol{\lambda}^*\right) \\ \lambda_i^* g_i\left(\mathbf{x}^*\right) = 0, i = 1, 2, \ldots, m \end{gathered}$$

Proof. We have

$$q\left(\lambda^*\right) = \min_{\mathbf{x} \in X} L\left(\mathbf{x}, \lambda^*\right) \leq L\left(\mathbf{x}^*, \lambda^*\right) = f\left(\mathbf{x}^*\right) + \sum_{i=1}^{m} \lambda_i^* g_i\left(\mathbf{x}^*\right) \leq f\left(\mathbf{x}^*\right)\,.$$

By strong duality, $\mathbf{x}^*, \boldsymbol{\lambda}^*$ are optimal iff $f\left(\mathbf{x}^*\right) = q\left(\boldsymbol{\lambda}^*\right)$. This is equivalent to $\min_{\mathbf{x} \in X} L\left(\mathbf{x}, \boldsymbol{\lambda}^*\right) = L\left(\mathbf{x}^*, \boldsymbol{\lambda}^*\right)$, and $\sum_{i=1}^{m} \lambda_i^* g_i\left(\mathbf{x}^*\right) = 0$, which in turn is equivalent to the slackness conditions in the theorem. □

We conclude these results with a more general duality theorem including convex affine inequality and equality contraints.

Theorem (General Strong Duality Theorem)**.** *Consider the optimization problem*

$$\begin{aligned} f^* = \min\ & f(\mathbf{x}) \\ s.t.\ g_i(\mathbf{x}) \leq 0, \quad & i = 1, 2, \ldots, m \\ h_j(\mathbf{x}) \leq 0, \quad & j = 1, 2, \ldots, p \\ s_k(\mathbf{x}) = 0, \quad & k = 1, 2, \ldots, q \\ \mathbf{x} \in X\,, & \end{aligned}$$

where X is a convex set and $f, g_i, i = 1, 2, \ldots, m$ are convex functions over X. The functions h_j, s_k are affine functions. Suppose that there exists $\hat{\mathbf{x}} \in \operatorname{int}(X)$ for which $g_i(\hat{\mathbf{x}}) < 0$, $h_j(\hat{\mathbf{x}}) \leq 0$, and $s_k(\hat{\mathbf{x}}) = 0$. Then if the problem has a finite optimal value, then the optimal value of the dual problem

$$q^* = \max\{q(\boldsymbol{\lambda}, \boldsymbol{\eta}, \boldsymbol{\mu}) : (\boldsymbol{\lambda}, \boldsymbol{\eta}, \boldsymbol{\mu}) \in \operatorname{dom}(q)\}$$

where

$$q(\boldsymbol{\lambda}, \boldsymbol{\eta}, \boldsymbol{\mu}) = \min_{\mathbf{x} \in X}\left[f(\mathbf{x}) + \sum_{i=1}^{m} \lambda_i g_i(\mathbf{x}) + \sum_{j=1}^{p} \eta_j h_j(\mathbf{x}) + \sum_{k=1}^{q} \mu_k s_k(\mathbf{x})\right]$$

is attained, and $f^* = q^*$.

Example. Consider the problem

$$
\begin{aligned}
\min x_1^3 + x_2^3 \\
x_1 + x_2 \geq 1 \\
x_1, x_2 \geq 0 \,.
\end{aligned}
$$

We will show that a problem can have different dual formulations with different duality gaps. It depends on our choice of X. Through the usual KKT conditions, we can easily find that $\left(\frac{1}{2}, \frac{1}{2}\right)$ is the optimal solution of the primal problem with an optimal value $f^* = \frac{1}{4}$. A first dual problem is constructed by taking $X = \{(x_1, x_2) : x_1, x_2 \geq 0\}$. The primal problem is $\min \left\{x_1^3 + x_2^3 : x_1 + x_2 \geq 1, (x_1, x_2) \in X\right\}$. Strong duality holds for the problem and hence in particular $q^* = \frac{1}{4}$. A second dual is constructed by taking $X = \mathbb{R}^2$. In this case, the objective function is not convex, implying that strong duality is not necessarily satisfied. In this case, the Lagrangian is given by

$$
L\left(x_1, x_2, \lambda, \eta_1, \eta_2\right) = x_1^3 + x_2^3 - \lambda\left(x_1 + x_2 - 1\right) - \eta_1 x_1 - \eta_2 x_2 \,.
$$

In this case, $q\left(\lambda, \eta_1, \eta_2\right) = -\infty$ for all $\left(\lambda, \eta_1, \eta_2\right) \Rightarrow q^* = -\infty$, and the duality gap is infinite. It is important to make a good choice of X to obtain useful information about the solution of the primal problem.

Three Important Examples of Duality Use

Linear Programming

Consider the linear programming problem

$$
\begin{aligned}
\min \quad & \mathbf{c}^T \mathbf{x} \\
\text{s.t.} \quad & \mathbf{A}\mathbf{x} \leq \mathbf{b},
\end{aligned}
$$

where $\mathbf{c} \in \mathbb{R}^n$, $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$. We assume that the problem is feasible, implying that strong duality holds. The Lagrangian is given by

$$
L(\mathbf{x}, \boldsymbol{\lambda}) = \mathbf{c}^T \mathbf{x} + \boldsymbol{\lambda}^T (\mathbf{A}\mathbf{x} - \mathbf{b}) = \left(\mathbf{c} + \mathbf{A}^T \boldsymbol{\lambda}\right)^T \mathbf{x} - \mathbf{b}^T \boldsymbol{\lambda}\,,
$$

and the dual objective function is

$$
q(\boldsymbol{\lambda}) = \min_{\mathbf{x} \in \mathbb{R}^n} L(\mathbf{x}, \boldsymbol{\lambda}) = \min_{\mathbf{x} \in \mathbb{R}^n} \left(\mathbf{c} + \mathbf{A}^T \lambda\right)^T \mathbf{x} - \mathbf{b}^T \boldsymbol{\lambda} = \begin{cases} -\mathbf{b}^T \boldsymbol{\lambda} & \mathbf{c} + \mathbf{A}^T \boldsymbol{\lambda} = 0 \\ -\infty & \text{else.} \end{cases}
$$

Therefore, the dual problem is formulated as

$$
\begin{aligned}
\max \quad & -\mathbf{b}^T \boldsymbol{\lambda} \\
\text{s.t.} \quad & \mathbf{A}^T \boldsymbol{\lambda} = -\mathbf{c} \\
& \boldsymbol{\lambda} \geq \mathbf{0}\,.
\end{aligned}
$$

Strictly Convex Quadratic Programming

Consider the strictly convex quadratic programming problem

$$\begin{array}{ll} \min & \mathbf{x}^T\mathbf{Q}\mathbf{x} + 2\mathbf{f}^T\mathbf{x} \\ \text{s.t.} & \mathbf{A}\mathbf{x} \leq \mathbf{b}\,, \end{array}$$

where $\mathbf{Q} \in \mathbb{R}^{n\times n}$ positive definite, $\mathbf{f} \in \mathbb{R}^n$, $\mathbf{A} \in \mathbb{R}^{m\times n}$, and $\mathbf{b} \in \mathbb{R}^m$. The Lagrangian (recall $\boldsymbol{\lambda} \in \mathbb{R}^m_+$) is given by:

$$L(\mathbf{x}, \boldsymbol{\lambda}) = \mathbf{x}^T\mathbf{Q}\mathbf{x} + 2\mathbf{f}^T\mathbf{x} + 2\boldsymbol{\lambda}^T(\mathbf{A}\mathbf{x} - \mathbf{b}) = \mathbf{x}^T\mathbf{Q}\mathbf{x} + 2\left(\mathbf{A}^T\boldsymbol{\lambda} + \mathbf{f}\right)^T\mathbf{x} - 2\mathbf{b}^T\boldsymbol{\lambda}\,.$$

The minimizer of the Lagrangian is attained at $\mathbf{x}^* = -\mathbf{Q}^{-1}\left(\mathbf{f} + \mathbf{A}^T\boldsymbol{\lambda}\right)$. With this, we work over the dual objective,

$$\begin{aligned} q(\boldsymbol{\lambda}) &= L\left(\mathbf{x}^*, \boldsymbol{\lambda}\right) \\ &= \left(\mathbf{f} + \mathbf{A}^T\boldsymbol{\lambda}\right)^T\mathbf{Q}^{-1}\mathbf{Q}\mathbf{Q}^{-1}\left(\mathbf{f} + \mathbf{A}^T\boldsymbol{\lambda}\right) - 2\left(\mathbf{f} + \mathbf{A}^T\boldsymbol{\lambda}\right)^T\mathbf{Q}^{-1}\left(\mathbf{f} + \mathbf{A}^T\boldsymbol{\lambda}\right) - 2\mathbf{b}^T\boldsymbol{\lambda} \\ &= -\left(\mathbf{f} + \mathbf{A}^T\boldsymbol{\lambda}\right)^T\mathbf{Q}^{-1}\left(\mathbf{f} + \mathbf{A}^T\boldsymbol{\lambda}\right) - 2\mathbf{b}^T\boldsymbol{\lambda} \\ &= -\boldsymbol{\lambda}^T\mathbf{A}\mathbf{Q}^{-1}\mathbf{A}^T\boldsymbol{\lambda} - 2\mathbf{f}^T\mathbf{Q}^{-1}\mathbf{A}^T\boldsymbol{\lambda} - \mathbf{f}^T\mathbf{Q}^{-1}\mathbf{f} - 2\mathbf{b}^T\boldsymbol{\lambda} \\ &= -\boldsymbol{\lambda}^T\mathbf{A}\mathbf{Q}^{-1}\mathbf{A}^T\boldsymbol{\lambda} - 2\left(\mathbf{A}\mathbf{Q}^{-1}\mathbf{f} + \mathbf{b}\right)^T\boldsymbol{\lambda} - \mathbf{f}^T\mathbf{Q}^{-1}\mathbf{f}\,. \end{aligned}$$

While this might look complicated, the resulting dual problem $\max\{q(\boldsymbol{\lambda}) : \boldsymbol{\lambda} \geq 0\}$ it is in fact another convex optimization problem, with a simpler feasible set than the primal problem.

Computing the Orthogonal Projection onto the Unit Simplex

Given a vector $\mathbf{y} \in \mathbb{R}^n$, we would like to compute the orthogonal projection of the vector $\mathbf{y}$ onto Δ_n. The corresponding optimization problem is

$$\begin{array}{ll} \min & \|\mathbf{x} - \mathbf{y}\|^2 \\ \text{s.t.} & \mathbf{e}^T\mathbf{x} = 1 \\ & \quad\mathbf{x} \geq 0\,. \end{array}$$

We will associate a Lagrange multiplier $\lambda \in \mathbb{R}$ to the linear equality constraint $\mathbf{e}^T\mathbf{x} = 1$ and obtain the Lagrangian function

$$\begin{aligned} L(\mathbf{x}, \lambda) &= \|\mathbf{x} - \mathbf{y}\|^2 + 2\lambda\left(\mathbf{e}^T\mathbf{x} - 1\right) = \|\mathbf{x}\|^2 - 2(\mathbf{y} - \lambda\mathbf{e})^T\mathbf{x} + \|\mathbf{y}\|^2 - 2\lambda \\ &= \sum_{j=1}^{n}\left(x_j^2 - 2\left(y_j - \lambda\right)x_j\right) + \|\mathbf{y}\|^2 - 2\lambda\,. \end{aligned}$$

The arising problem is therefore separable with respect to the variables x_j and hence the optimal x_j is the solution to the one-dimensional problem

$$\min_{x_j \geq 0}\left[x_j^2 - 2\left(y_j - \lambda\right)x_j\right]$$

The optimal solution to the above problem is given by

$$x_j = \begin{cases} y_j - \lambda, & y_j \geq \lambda \\ 0 & \text{else} \end{cases} = \left[y_j - \lambda\right]_+$$

and the optimal value is $-\left[y_j - \lambda\right]_+^2$. The dual problem is therefore

$$\max_{\lambda \in \mathbb{R}} \left\{ g(\lambda) \equiv -\sum_{j=1}^{n} \left[y_j - \lambda\right]_+^2 - 2\lambda + \|\mathbf{y}\|^2 \right\}$$

By the basic properties of dual problems, the dual objective function is concave. In order to actually solve the dual problem, we note that

$$\lim_{\lambda \to \infty} g(\lambda) = \lim_{\lambda \to \infty} g(\lambda) = -\infty$$

Therefore, since $-g$ is a coercive and differentiable function, it follows that there exists an optimal solution to the dual problem attained at a point λ in which

$$g'(\lambda) = 0\,,$$

meaning that

$$\sum_{j=1}^{n} \left[y_j - \lambda\right]_+ = 1\,.$$

The function $h(\lambda) = \sum_{j=1}^{n} \left[y_j - \lambda\right]_+ - 1$ is a nonincreasing function over $\mathbb{R}$ and is in fact strictly decreasing over $\left(-\infty, \max_j y_j\right]$. In addition, by denoting $y_{\max} = \max_{j=1,2,\ldots,n} y_j$, and $y_{\min} = \min_{j=1,2,\ldots,n} y_j$, we have

$$\begin{aligned} h\left(y_{\max}\right) &= -1 \\ h\left(y_{\min} - \frac{2}{n}\right) &= \sum_{j=1}^{n} y_j - n y_{\min} + 2 - 1 > 0\,, \end{aligned}$$

and we can therefore invoke a bisection procedure to find the unique root λ of the function h over the interval $\left[y_{\min} - \frac{2}{n}, y_{\max}\right]$ and then define $P_{\Delta_n}(\mathbf{y}) = [\mathbf{y} - \lambda \mathbf{e}]_+$.

Dual Algorithms

Duality relies in the fact that no matter the nature of the objective function f (convex or not), the dual problem will always involve the maximisation of a concave function. However, this does not necessarily implies that there will be an explicit expression for the solutions of the dual problem. Thus, iterative method have been developed focused on the existing relations between the primal and dual variables.

Augmented Lagrangian

Let us consider the constrained optimisation problem

$$\begin{aligned} \min_{\mathbf{x}\in X} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & A\mathbf{x} = b \end{aligned}$$

with associated dual problem

$$\max_{\boldsymbol{\mu}} q(\boldsymbol{\mu}).$$

The main difficulty encountered while solving the dual problem is the possible nonsmoothness of the dual objective. One approach to deal with this fact is use a very common iterative algorithm for nonsmooth optimisation called the *proximal point method*, which is described through the following generic iteration:

$$\boldsymbol{\mu}^{k+1} \leftarrow \operatorname*{argmax}_{\boldsymbol{\mu}} q(\boldsymbol{\mu}) - \frac{1}{2\alpha_k}\|\boldsymbol{\mu} - \boldsymbol{\mu}^k\|^2$$

with $\alpha_k > 0$. Using the definition of the dual function, we have

$$\boldsymbol{\mu}^{k+1} = \operatorname*{argmax}_{\boldsymbol{\mu}} \min_{\mathbf{x}} \left\{ f(\mathbf{x}) + \boldsymbol{\mu}^\top (A\mathbf{x} - b) - \frac{1}{2\alpha_k}\|\boldsymbol{\mu} - \boldsymbol{\mu}^k\|^2 \right\}$$

This is a *saddle point problem* on $(\mathbf{x}, \boldsymbol{\mu})$. When f is convex, we have that the objective function is convex on $\mathbf{x}$ and strongly concave on $\boldsymbol{\mu}$, and therefore we can interchange the order of optimisation, i.e., now we consider the problem

$$\min_{\mathbf{x}} \max_{\boldsymbol{\mu}} \left\{ f(\mathbf{x}) + \boldsymbol{\mu}^\top (A\mathbf{x} - b) - \frac{1}{2\alpha_k}\|\boldsymbol{\mu} - \boldsymbol{\mu}^k\|^2 \right\}$$

Solving for $\boldsymbol{\mu}$, we obtained that the optimal solution is

$$\boldsymbol{\mu} = \boldsymbol{\mu}^k + \alpha_k (A\mathbf{x} - b),$$

substituting this in the original problem we obtain

$$\min_{\mathbf{x}\in X} f(\mathbf{x}) + (\boldsymbol{\mu}^k + \alpha_k(A\mathbf{x} - b))^\top (A\mathbf{x} - b) - \frac{\alpha_k^2}{2\alpha_k}\|A\mathbf{x} - b\|^2.$$

After some manipulation we obtain the following problem

$$\operatorname*{argmin}_{\mathbf{x}\in X} f(\mathbf{x}) + (\boldsymbol{\mu}^k)^\top (A\mathbf{x} - b) + \frac{\alpha_k}{2}\|A\mathbf{x} - b\|^2 =: L^A_{\alpha_k}(\mathbf{x}, \boldsymbol{\mu}) \qquad \text{(AL-P)}$$

The function $L^A_{\alpha_k}$ is called the *augmented Lagrangian*, since it consists of the ordinary Lagrangian function added to a quadratic term that penalizes the violation of the equality constraint $A\mathbf{x} = b$. A generic iteration of the algorithm consists in three main steps

1. minimisation of the augmented Lagrangian L^A for $\mathbf{x}$ with fixed $\boldsymbol{\mu}$ and parameter α_k
2. Update of the dual variable $\boldsymbol{\mu}$

3. Update of the parameter α_k

The *augmented Lagrangian method* converges even in the case of constant parameter $\alpha_k = \alpha$. However, some heuristics can be used to improve its performance, as we show in the following algorithm:

Algorithm 1: The Augmented Lagrangian Method

Initialization: $\boldsymbol{\mu}^0$, initial parameter $\alpha_1 > 0$ and $\delta_1 = +\infty$
Initialization: Set the parameters $\eta \in (0, 1)$ and $\gamma > 1$
General Step: for any $k = 0, 1, 2, \ldots$ execute the following steps:

```
Set x^k by solving problem AL-P
Set δ = ||Ax − b||^2
if δ < ηδ_k then
    μ^{k+1} ← μ^k + α_k(Ax^k − b)   // Acceptable improvement in the feasibility; update μ
    α_{k+1} ← α_k
    δ_{k+1} ← δ
else
    μ^{k+1} ← μ^k                   // Insufficient improvement in feasibility; do not accept μ
    α_{k+1} ← γα_k                  // increase the value of α
    δ_{k+1} ← δ_k
```

Alternating Direction Method of Multipliers

This method is targeted to problems of the form

$$\begin{aligned} \min_{\mathbf{x},\mathbf{z}} \quad & f(\mathbf{x}) + g(\mathbf{z}) \\ \text{s.t.} \quad & A\mathbf{x} + B\mathbf{z} = c \end{aligned} \tag{ADMM-P}$$

with $\mathbf{x} \in X$ and $\mathbf{z} \in Z$, where X, Z are closed and convex sets.

Let us consider the augmented Lagrangian associated to problem ADMM-P

$$L_\alpha^A(\mathbf{x}, \mathbf{z}, \boldsymbol{\mu}) = f(\mathbf{x}) + g(\mathbf{z}) + \boldsymbol{\mu}^\top (A\mathbf{x} + B\mathbf{z} - c) + \frac{\alpha}{2}\|A\mathbf{x} + B\mathbf{z} - c\|^2$$

The Alternating Direction Method of Multipliers or ADMM consists mainly on the following steps:

$$\begin{aligned} \mathbf{x}^{k+1} &= \operatorname*{argmin}_{\mathbf{x} \in X} L_{\alpha_k}^A(\mathbf{x}, \mathbf{z}^k, \boldsymbol{\mu}) \\ \mathbf{z}^{k+1} &= \operatorname*{argmin}_{\mathbf{z} \in Z} L_{\alpha_k}^A(\mathbf{x}^{k+1}, \mathbf{z}, \boldsymbol{\mu}^k) \\ \boldsymbol{\mu}^{k+1} &= \boldsymbol{\mu}^k + \alpha_k \left(A\mathbf{x}^{k+1} + B\mathbf{z}^{k+1} - c \right) \end{aligned}$$

The main feature of ADMM is the independent optimisation of each primal variable. In some contexts, optimising the primal variables separately results advantageous in comparison with usual methods of multipliers like the augmented Lagrangian method.

We end this chapter by showing an example of how to rewrite a usual constrained optimisation problem to fit the structure of problem ADMM-P.

Example Let us consider the following problem known as *Linear Quadratic Programming*

$$\begin{aligned} \min_{\mathbf{x}} \quad & \mathbf{c}^\top \mathbf{x} + \frac{1}{2}\mathbf{x}^\top Q \mathbf{x} \\ \text{s.t.} \quad & A\mathbf{x} = b \\ & \ell \leq \mathbf{x} \leq u \end{aligned}$$

where $Q \succ 0$ and ℓ, u vectors of lower and upper bounds on the components of $\mathbf{x}$, respectively.

To reformulate this problem as ADMM-P, we introduce a new variable $\mathbf{z}$ such that

$$\begin{aligned} \min_{\mathbf{x}} \quad & \mathbf{c}^\top \mathbf{x} + \frac{1}{2}\mathbf{x}^\top Q \mathbf{x} \\ \text{s.t.} \quad & A\mathbf{x} = b \\ & \mathbf{z} = \mathbf{x} \\ & \ell \leq \mathbf{z} \leq u \end{aligned}$$

Now, we consider the following augmented Lagrangian

$$L_\alpha^A(\mathbf{x}, \mathbf{z}, \boldsymbol{\mu}) = \mathbf{c}^\top \mathbf{x} + \frac{1}{2}\mathbf{x}^\top Q \mathbf{x} + \boldsymbol{\mu}^\top (\mathbf{x} - \mathbf{z}) + \frac{\alpha}{2}\|\mathbf{z} - \mathbf{x}\|^2$$

and we choose to enforce the constraints $A\mathbf{x} = b$ and $\ell \leq \mathbf{z} \leq u$ explicitly. The ADMM updates are then given by:

$$\begin{aligned} \mathbf{x}^{k+1} &= \operatorname*{argmin}_{\mathbf{x}} L_{\alpha_k}^A(\mathbf{x}, \mathbf{z}^k, \boldsymbol{\mu}^k) \text{ subject to } A\mathbf{x} = b \\ \mathbf{z}^{k+1} &= \operatorname*{argmin}_{\mathbf{z}} L_{\alpha_k}^A(\mathbf{x}^{k+1}, \mathbf{z}, \boldsymbol{\mu}^k) \text{ subject to } \ell \leq \mathbf{z} \leq u \\ \boldsymbol{\mu}^{k+1} &= \boldsymbol{\mu}^k + \alpha_k(\mathbf{x}^{k+1} - \mathbf{z}^{k+1}) \end{aligned}$$

The $\mathbf{x}$ update can be obtained by using KKT-conditions for the equality constrained problem. The $\mathbf{z}$ update has a closed form solution, since the projection operator for the box-constraints can be explicitly stated.